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Efficient Clustering Based Routing for Energy Management in Wireless Sensor Network-Assisted Internet of Things

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Abstract: Wireless sensor networks (WSNs) play a huge part in arising innovations like smart applications, the Internet of Things, and numerous self-designed, independent applications. Energy exhaustion and efficient energy consumption are principal issues in wireless sensor networks. Energy is a significant and valuable asset of sensor nodes; early energy depletion ultimately leads to a shorter network lifetime and the replacement of sensor nodes. This research proposes a novel Power-Efficient Cluster-based Routing (PECR) algorithm. It takes in clustering using K-Means, the arrangement of Cluster Heads (CHs) and a Main Cluster Head (MCH), the optimal route choice, communication in light of the energy utilization model, cluster heads, and main cluster head alternation based on residual energy and relative location. PECR decreases traffic overburden, restricts energy usage, and at last, expands the network lifetime. Sensor nodes sense the information and transmit traffic to a Base Station (BS) through a legitimate channel. The results confirm it decreases the traffic overhead and effectively utilizes the energy assets. The simulation results show that PECR's performance is 44% more improved than LEACH, EC, EECRP, and EECA algorithms. It is suitable for networks that require a stretched life.

Keywords: cluster-based; internet of things (IoT); routing protocols; wireless sensor network (WSN)

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1. Introduction

The Internet of Things is one of the growing advancements of the 5G establishment [1]. It is helpful in various pieces of the lifecycle, for instance, industrialized control, sensing environments, and military areas [2]. The IoT is the correspondence between actual devices, methods for conveyance, construction slabs, and various objects that comprise installed devices, programs, actuators, and sensing tools, and the rundown continues [3]. The IoT enables smart objects to accumulate and exchange information for different purposes [4]. Regularly, WSN gives an illustration of the IoT network. It is an immense viscosity of devices with detection capabilities that are introduced to detect and screen the encompassing state of an IoT system. Each device is required to communicate accumulated information to a sink/BS through multiple midpoints. In the WSN-assisted IoT, the sensor nodes have restricted resources according to different points of view, for instance, storage, processing, energy, and so forth. In light of the fleeting power of sensors and the tremendous expense of sensor node substitution, subbing the sensor node has poor value. In this manner, expanding the network's existence time and energy resource management throughout the network is the main point while filtering the execution of routing protocols in WSN. In this situation, overwhelming routing protocols are expected to extend framework life and achieve the improved utilization of power assets.

This exploration study investigates energy management, which is a significant quality of service parameter in the WSN-assisted IoT. Besides communications, a sensor node's energy is consumed in other activities as well such as logging, processing, sensing, and a few more [5]. If a sensor node runs out of energy and is unable to send or receive packets, a part

of the network coverage area may also become disconnected. Wireless sensor node usage may be found in hostile scenarios as well, which makes node replacement and recharging almost impossible. This incorporates the proficient utilization of restricted energy assets of nodes present in WSN to prolong the network lifetime. The proposed algorithm is power-effective cluster-based routing (PECR). It focuses on managing energy assets and at last broadens the lifetime of the WSN-assisted IoT. Additionally, this work incorporates the investigation of previous cluster-based routing algorithms and their performance in dealing with energy assets.

Background Studies

Cluster-based routing protocols go about their responsibilities on the network layer; this layer provides a link between the upper and lower layers of the network. A few keys and updated versions are examined here. LEACH is the elementary tiered, self-shaping, routing protocol planned for use for WSN. In LEACH, the rotation of the CH is the probability-based choice; in this situation, the choice of a node with low energy such as a CH is conceivable. It causes an immense volume of energy exhaustion [6]. LEACH-C [7] uses a focal clustering algorithm. The immense amount of transmission of information toward the beginning rounds causes a huge overhead. Its performance varies from LEACH if the BS is exclusively situated. The PEGASIS protocol utilizes the greedy algorithm to form a chain of nodes [8].

A node needs more energy to send data to a distant neighbor. Experiencing the same thing, its energy will drain sooner than others. TL-LEACH is very much similar to LEACH. A CH gathers and joins all information and uses multi-hop transmission to the BS [9]. M-LEACH tracks down the best, most limited way to forward data to the sink involving a halfway node to forestall the quick exhaustion of power [10]. V-LEACH contains CHs as well as vice CHs too. However, this happens just when CH kicks the bucket because of energy dissemination [11]. In LEACH-B, the optimum number of CHs was viewed as significant as per the subsequent CH calculation [12]. Although LEACH-B has no unified control, it has a dispersed algorithm, wherein nodes settle on independent decisions. EC is a dispersed clustering algorithm that specifies a suitable size for a cluster, as well as the positions on the node's distance to sink [13]. GSA is a stochastic population-based meta-heuristic algorithm for the near-optimal positioning of the BS in heterogeneous two-tiered WSNs [14,15]. It finds a better location when compared to the PSO algorithm and the exhaustive search, but the problem of cluster formation is not addressed. GGSA is an extension of GSA for resolving the data clustering problem, so data gathered from a cluster is unlike data from other clusters [16], but it does not address the energy efficiency constraints of WSNs.

E-LEACH provides the determination method of CH that depends on the nodes' energy level [17]. P-LEACH is a blend of PEGASIS and LEACH. It joins the properties of the two conventions and gives the method of chain creation inside a cluster [18]. In HEED, the CH choice depends on leftover energy [19]. H-LEACH chooses CHs utilizing the average and residual energy of the nodes [20]. The GEEC algorithm provides a CH determination process that depends on the transformative game model. It shows significant improvement over LEACH-C and LEACH [21]. EECRP provides CH determination that depends on the place of the centroid. In this manner, the inclusion of the entire network is amplified [22]. In the PEAL algorithm, the sensing field is partitioned into zones. The transmission mode among the CHs and the BS depends on the zone partition. However, a lot of control messages can cause fast energy consumption [23]. In the EECR algorithm, the cost method given the distance and energy is used for multi-hop transmission among CHs; it eventually safeguards from energy problems [24]. In an elite group routing algorithm, the CH choice is restricted to a couple of nodes that have high energy [25]. However, CHs have direct admittance to the sink.

In MLSEEP, the node with the most elevated energy level is chosen as the CH. The CH sends its information to the BS utilizing multi-hop transmission because of fuzzy logic [26].

However, every node does not have an equivalent chance to become a CH. In the UCRA algorithm, the CHs are chosen in each circle from part nodes by utilizing clocks while considering the present residual energy situation in the circle field [27]. In the FBUCA algorithm, sensor nodes are sent haphazardly in a three-layer framework. The CHs are chosen by the fuzzy calculation. The transmission cycle happens in a multi-hop mode from layer 0 to layer 2 [28]. In EECA, K-Means grouping is utilized, which depends on the energy and distance of nodes that further enhance the grouping technique [29]. In this algorithm, the BS is portable. In a couple of situations where the BS is outside the detecting field, the energy consumption rate might be higher. In the EECMHR algorithm, the CHs are picked haphazardly, and the nodes are combined in groups. The CHs straightforwardly communicate with the BS [30]. The CHs, which are far away from the BS, can drain their energy faster. In a study, the proposed algorithm gives 60% better results than LEACH on account of energy utilization and network life [31]. However, direct transmission between CHs and the BS can exhaust energy more quickly.

GMA proposed to create K mutually different clusters of sensing nodes so that each subset of sensors alone can cover the entire network, leading to maximum sleeping nodes and hence saving energy [32], but CH election is not taken into account. The LEACH-opt cluster addresses the improper cluster head selection election in LEACH by finding out the ideal cluster number while keeping in view the total energy already consumed [33]. Compared to LEACH, on average, the overall survival rate of the working nodes in the network increased by more than 30%. A central cluster algorithm based on an improved genetic algorithm (EGA) that finds appropriate numbers of CHs in networks is proposed [34]. It outperforms LEACH in a longer network lifetime measured in terms of the first node dying, half of the nodes dying, and the last node dying but does not consider residual energy while selecting CHs. The CAFR algorithm proposed cluster head selection based on a family relationship that is decided on two parameters, i.e., a node's remaining energy and communication cost [35]. The performance of the algorithm is judged on only two parameters: network operation life cycle and total data traffic. The BS is placed in the center of the network to achieve energy efficiency [36]; however, this solution does not apply to all possible scenarios of WSNs.

LEACH and its different variants are well-studied. They are attempts to enhance the overall performance of the WSN by expanding the lifetime of the WSN, decreasing the traffic overhead, increasing the packet delivery ratio, and balancing the energy utilities. Some techniques use the cluster head formation process and other CH rotation processes. Several divide the sensing field into two layers or multiple circular layers for cluster formation. Few of them use multi-hop transmission, while others use single-hop. Not a single calculation gives 100% productive energy execution. There are spaces to track down holes and tackle these energy management problems in the WSN-assisted IoT. The outline of the paper is given below. Section 2 provides materials and methods including the proposed algorithm in detail. Section 3 includes simulation results. Section 4 shows the discussion of the results. Section 5 concludes this research work.

2. Materials and Methods

This section elaborates on the methodology of the PECR algorithm in detail.

2.1. Initial Suppositions

A few assumptions are made at first in the design development of the PECR algorithm. The deployment of sensor nodes is random and static. The location of the BS is in the middle of the network. The nodes are location-aware. The PECR algorithm does not allow direct access from CHs to the BS. During the process of transmission, the traffic reaches the BS through the shortest path using rely-on-point MCH.

2.2. Initialization Phase

The message is sent by sensor nodes to the BS that incorporates the node's X and Y coordinates and ID addressing the energy level and location. The BS computes the distance to every node and stores data. The BS performs the K-Means [6] algorithm to create clusters. Each distinct color shows a separate cluster, as portrayed in Figure 1.

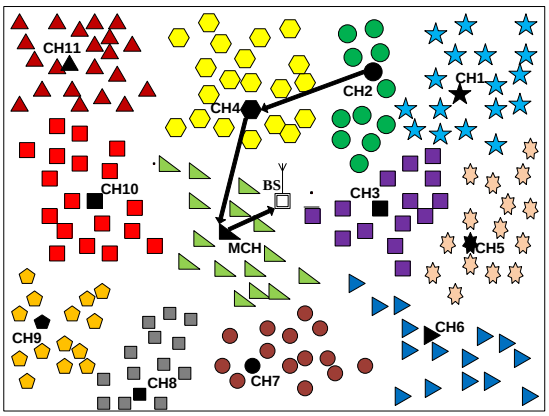

Figure 1. Optimal route-finding in the PECR algorithm.

First cluster head selection in each cluster: The BS examines the stored energy level and location of the nodes in the individual cluster and selects one node as the CH that has a minimum distance from the participant nodes and maximum energy. This process repeats for all clusters, and the BS stores this information. As Figure 1 shows, the CH is symbolized in a dark color in each cluster.

First main cluster head selections: The MCH is the relay point between all CHs and the BS. After selecting the CH in each cluster, the BS selects the first MCH. This process includes distance calculations from the BS to each CH. After this, the BS selects one CH which is the MCH with the least distance from the BS, as Figure 1 portrays.

Feedback message from the base station to the sensor nodes: Once the CH and MCH are selected in each cluster, the BS sends a feedback message to each node. It includes the threshold energy and location of its corresponding CH and MCH. This information is useful in the transmission phase while searching for an optimum path.

2.3. Transmission Phase

In the PECR algorithm, the transmission phase involves communication from the sensor nodes to the BS on different levels and using different amounts of energy according to the proposed energy consumption model given below in detail.

The energy consumption model is revised from [19] which shows (1) the transmission energy of a node for forwarding energy information and location to the BS and (2) for the reception of feedback messages from the BS in the initialization phase. Equation (3) shows the calculation of the threshold distance.

$$E_{Tx}(l,d) = \begin{cases} l_{ength} \times (e_t + \epsilon_{fs} \times d^2), if\ d \leq d_{th} \\ l_{ength} \times (e_t + \epsilon_{amp} \times d^4), if\ d > d_{th} \end{cases} \quad (1)$$

$$E_{Rx}(l) = l_{ength} * e_r \quad (2)$$

where,

l = length of the data packet.

d = the distance of the respective node.

e_t = the transmit energy.

e_r = the receiving energy.

ϵ_{fs} = the free space energy (energy consumption amplification factor of the free space model).

ϵ_{amp} = the transmit amplifier energy (energy consumption amplification factor of the multipath fading model).

d_{th} = the threshold distance.

The threshold distance d_{th} is calculated as,

$$d_{th} = \sqrt{\frac{\epsilon_{fs}}{\epsilon_{amp}}} \tag{3}$$

In the transmission phase, the data communication and energy utilization at each level is determined independently and is given underneath.

2.3.1. Data Communication and Energy Consumption from Nodes to Cluster Head

At first, the nodes collect the required information from the deployed field. The nodes execute minor processing and forward data to the CH utilizing the least energy as determined by the energy consumption model specified below in (4) and (5).

$$E_{Nodes_{Tx}} = \eta \times l_{ength} \times \left(e_t + \epsilon_{fs} \times d^2\right) \tag{4}$$

$$E_{Nodes_{Rx}} = \eta \times l_{ength} \times e_r \tag{5}$$

Here, η shows the sum of the nodes (excluding the CH) within one cluster.

2.3.2. Data Communication and Energy Consumption from Cluster Head to Main Cluster Head

The CH collects data from the nodes and forwards them in a compacted form to the MCH through at least one jump utilizing the minimum amount of energy that is assessed by Equations (6) and (7).

$$E_{CH_{Tx}} = \Lambda \times l_{ength} \times \left(e_t + \epsilon_{fs} \times d^2\right) \tag{6}$$

$$E_{CH_{Rx}} = \Lambda \times l_{ength} \times \left(e_r + E_{agg}\right) \tag{7}$$

Here, Λ shows the sum of the nodes under one CH.

In a multi-hop scenario, the CH finds the most reasonable way to reach the MCH. It can be found in the proposed route determination scheme given below.

2.3.3. Selection of Optimum Route

In the PECR calculation, if CHs are situated far away, they do not advance collected data straightforwardly to the MCH. In the multi-hop transmission mode, a CH sends route demands to CHs situated in a forward way to the MCH, and the BS and chooses one which takes less time to reply. The minimum reply time indicates the minimum distance among the CHs. The selected CH then receives the data, combines them with its own, and forwards them in a compacted form to the MCH as displayed in Figure 1.

2.3.4. Data Communication and Energy Consumption at Main Cluster Head to Base Station

The MCH collects data from its part nodes and aggregates them with data received from all CHs. Equations (8) and (9) calculate the energy utilized in this process determined by the energy consumption model.

$$E_{MCH_{Tx}} = \mathfrak{w} \times l_{ength} \times \left(e_t + \epsilon_{fs} \times d^2\right) \tag{8}$$

$$E_{MCH_{Rx}} = \mathfrak{w} \times l_{ength} \times \left(e_r + E_{agg}\right) \tag{9}$$

Here, $\mathfrak{w}$ is the total number of CHs in the network.

The MCH forwards the compacted data to the BS using the minimum energy that is calculated as given in (10).

$$E_{MCH-BS_{Tx}} = \mathbb{N} \times l_{ength} \times \left(e_t + \epsilon_{fs} \times d^2\right) \tag{10}$$

Here, $\mathbb{N}$ shows the sum of sensors in the deployed field.

2.4. Rotation Phase

In WSN, energy is a significant constraint. The CHs and the MCH that are chosen at first by the BS do not continue their position all through their lifetime. Therefore, after the fulfillment of one round, the CHs and the MCH rotate to divide the energy consumption burden on all sensor nodes.

2.4.1. Inter-Cluster Selection Process

In this process, a new CH is chosen by the former CH. This depicts the self-adjusting nature of the network. The current CH checks the remaining energy and position of each node within a cluster. The energy level of the new CH is maximum with the least relative distance from the other nodes in a cluster. The same process repeats for the new MCH rotation within a cluster. After rotating, the MCH and CHs share path information with the other nodes and the BS. Based on this, the nodes bring their directing table up-to-date. This revolution interaction rehashes in every round to reduce the energy consumption and prior expiration of the nodes.

2.4.2. Declaration of Dead Nodes

Energy is an important parameter to keep the network functional. Therefore, a node is acknowledged as dead if it has less energy than the threshold energy level (calculated by the BS in the initialization phase). The CH maintains a record of the dead nodes, and these nodes do not take part in the data transmission process.

2.5. Workflow of Power-Efficient Cluster-Based Routing Algorithm

Figure 2 below illustrates the flow of the PECR algorithm.

The main focus of this study is to find the factors behind energy exhaustion and the efficient use of energy assets. As discussed, in previous algorithms, there are a lot of loopholes that drain sensor nodes' energy more quickly. In this study, the proposed methodology includes intermediate levels during the transmission phase. The first level is from the node to the CH, where transmission and reception energy consumption is estimated by considering the length of the data packet, the free space energy, and the distance of the respective node as given in Equations (4) and (5). At the second level (CH to MCH) along with the previous parameters, aggregation energy is also considered, as given in Equation (7). The third level (MCH to BS) is important in the proposed study. At this level, energy consumption is estimated using all previous parameters as given in Equations (8)–(10). In the proposed energy consumption model of the PECR algorithm, energy has a direct relation with distance (d). Therefore, a multi-hop scheme is designed to conserve the energy of each node. Additionally, there is no direct data transmission between the sensor nodes to the BS. In short, the PECR algorithm is focusing on solving the limitations of existing algorithms using a clustering scheme, CH and MCH functionalities, optimum path selection in a forward direction, the rotation process, and energy consumption estimation at each level of data transmission.

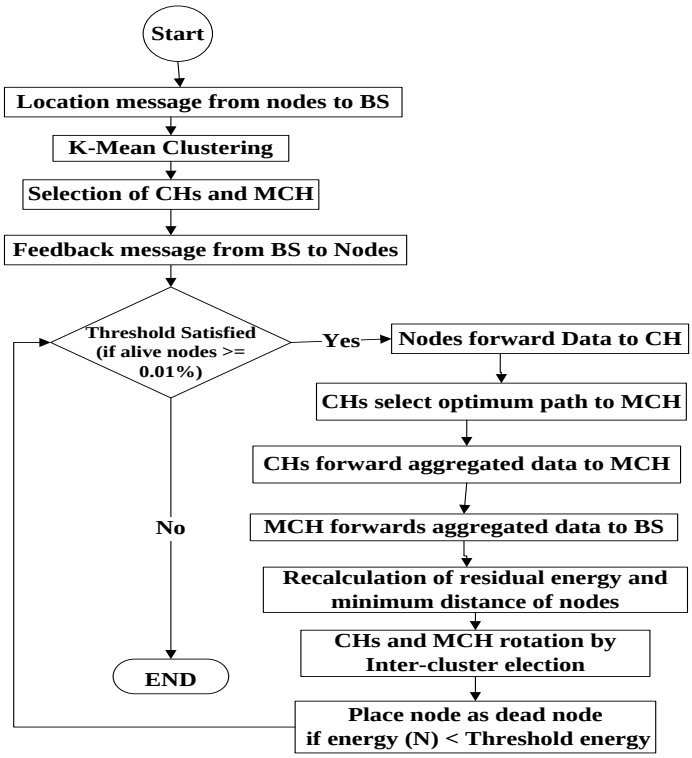

Figure 2. The flow of the PECR algorithm.

3. Results

This section includes the simulation setup, the parameters used in simulations, and detailed simulation results.

Simulation Setup

MATLAB is used for Simulations of PECR algorithm utilizing a couple of parameters given in Table 1.

Table 1. Parameters for simulations.

Factors	Values
Field's dimensions	200×200 m^2
E_{th} (Threshold energy)	0.000005 µJ/bit
I_e (initial energy)	0.5 J
ϵ_{fs}	10×10^{-12} J/bit/m^2
e_r (receiving energy)	50×10^{-9} J/bit
l (data length)	4000 bits
BS's locations	Inside (center)
e_t (transmission energy)	50×10^{-9} J/bit
d_{th} (threshold distance)	87 m
ϵ_{amp}	1.3×10^{-15} J/bit/m^4

In the proposed PECR algorithm, 500 low-power, stationary, and position-aware sensor nodes are arbitrarily installed in a 200×200 m^2 area as shown in Figure 3. The BS is located at the focal point of the field. The goal is the effective and balanced utilization of energy assets.

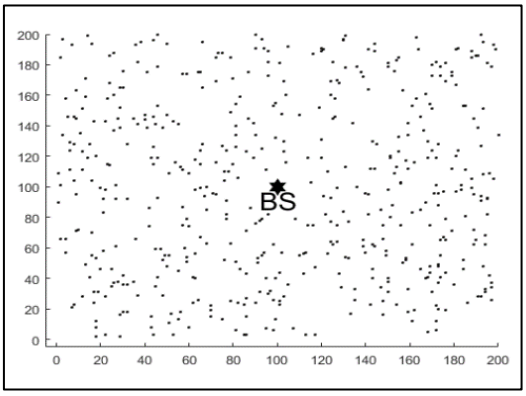

Figure 3. Nodes positioned in a 200 × 200 m^2 field.

After the arbitrary installation of nodes in the sensing field, the K-Means clustering algorithm (run by BS) frames the non-covering diverse clusters of the nodes as displayed in Figure 4.

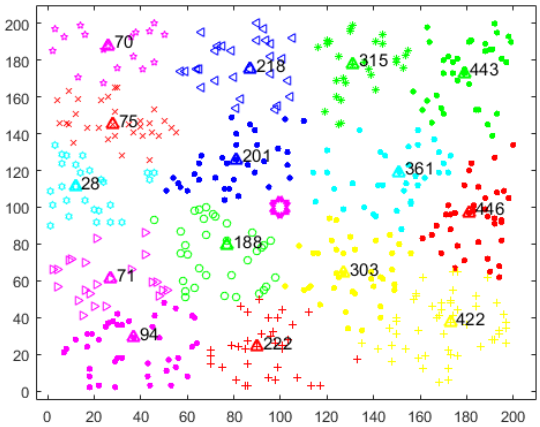

Figure 4. K-Means clustering in the PECR algorithm.

Each shape in one tone addresses a separate cluster. The star shape in the center addresses the BS. A three-sided shape addresses the CH in each cluster.

The PECR algorithm uses a multi-hop transmission mode from node to BS utilizing low energy assessed by the proposed energy consumption model.

Figure 5 depicts the competitive performance of the PECR algorithm in terms of energy usage against the number of rounds. Initially, after the establishment of the network at zero rounds, all nodes have maximum energy. The number of transmission rounds and the energy usage of the network have a direct correlation. Energy utilization increases with an increase in rounds. There is no sudden change in the curve. The energy utilization level progressively expands by up to 1800 rounds.

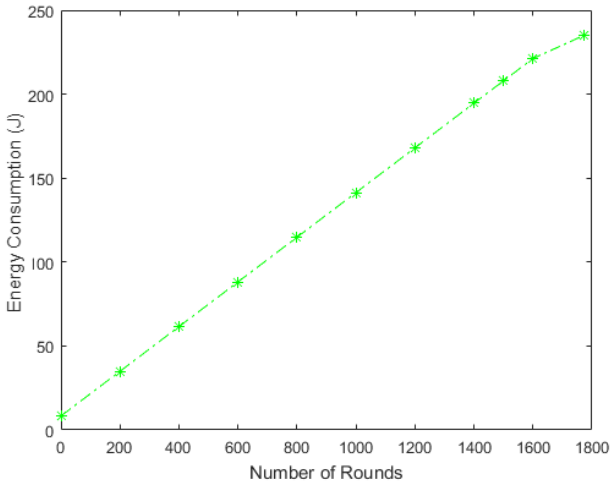

Figure 5. Total network energy usage.

4. Discussion

An evaluation of the PECR algorithm is needed to check its competitive results for a WSN-assisted IoT network. Four existing algorithms that have the same simulation parameters and environment are chosen to compare the results. The existing algorithms involved in this evaluation are LEACH [6], EC [13], EECRP [19], and EECA [27]. Then, the simulations are conducted utilizing similar factors and platforms for a fair comparison. The simulation results are analyzed based on three evaluation criteria to compare the throughput of the PECR and the other three algorithms.

The first evaluation criterion is the sum of alive nodes that demonstrate the sensor nodes participating in the data transmission effectively. The sum of alive nodes presents the absolute life period of the network which is the main imperative in routing protocols.

Figure 6 shows a huge contrast in the general number of alive nodes until the fruition of rounds. LEACH has the worst throughput among all because of a plenty of control messages and the absence of a middle point between the CHs and the BS. After 800 rounds, it does not have adequate alive nodes. EC has better execution when contrasted with LEACH, yet not as viable. In both EC and LEACH, the probability model does not ponder about leftover energy. There are a more noteworthy number of chances that it would choose a low-energy CH. This prompts fast energy depletion, and the nodes are not alive after 1000 rounds. EECRP works in a single-hop mode, which causes the energy exhaustion of the CHs. It does not continue after 1100 rounds. EECA has a more prominent number of rounds when contrasted with the past three calculations. However, it could not continue after 1300 rounds. The PECR algorithm outpaces it, as displayed in Figure 6. The initial node expires almost at the 800th round. Clustering and the multi-hop optimal way selection guarantee the balanced utilization of energy assets. It outperforms LEACH by 55%, EC by 52%, EECRP by 32%, and EECA by 26%.

Figure 7 portrays the second evaluation criterion, which is during 1200 rounds, the quantity of sent data packets is effectively obtained by the BS. LEACH has a data packet delivery ratio close to 3×105. It does not look at the leftover energy of the functional nodes while choosing the CHs, which causes quick energy exhaustion and fewer rounds. EC does not look at the distance all through the course of the midpoint determination, which means extra energy utilization and less successful transmission.

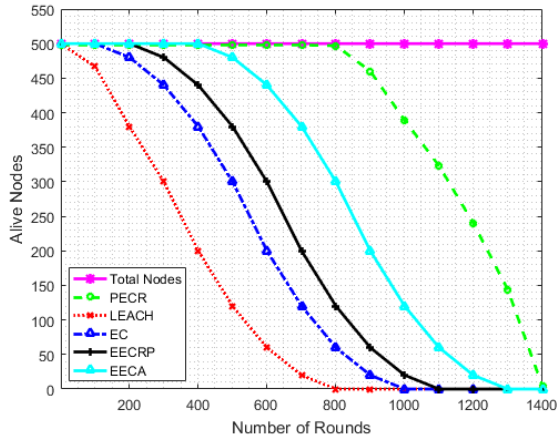

Figure 6. The overall sum of alive nodes.

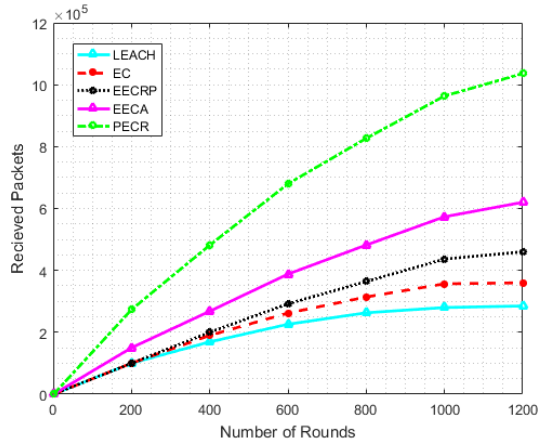

Figure 7. Packets received by the BS.

EC shows a 3.5 × 105 ratio. EECR somehow has better outcomes because of its protective procedure [5] and winds up to 4.5 × 105, while EECA winds up to 6.2 × 105.

Both EECR and EECA are poorer than the PECR as it consumes more energy in a single jump transmission. The PECR shows more effective performance, winding up to 10.5 × 105. This is because of the optimal multi-hop transmission mode and energy utilization methodology. The result portrays PECR as having 62.5% preferable performance over LEACH, 58% more than EC, 50% more than EECRP, and 35% more than EECA.

The third significant assessment metric is the remaining energy. Right away, all sensor nodes have an equivalent energy level that is 0.5 J, which is fixed in case there were no transmissions. As displayed in Figure 8, LEACH has a quick energy consumption level because of a probability-based model and all nodes become lapsed at just about 600 rounds. EC checks the proficient utilization of energy and the dynamic clusters. However, it does not ensure the position of the CHs. In this way, EC closes at right around 800 or a few more rounds. EECRP shows better execution because of clustering in light of the distance through the energy centroid. However, the CHs have direct correspondence with BS that closes EECRP at very nearly 900 transmission rounds, while EECA closes at just about 1000 transmission rounds. The PECR algorithm wins in terms of the management of energy utilization.

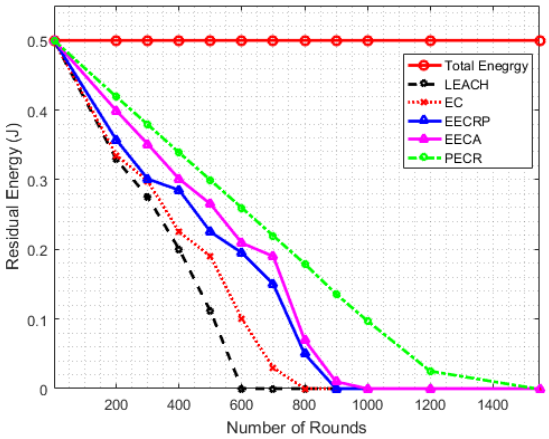

Figure 8. The residual energy of the WSN.

It is based on multi-hop transmission and an energy utilization model that reduces the significant distance transmission and immoderate utilization of restricted energy assets. Accordingly, the PECR algorithm covers extra transmission rounds with a comparative measure of starting energy. Figure 8 shows that the energy exhaustion pace of the PECR algorithm is 57% lesser than LEACH, 43% lesser than EC, 35% lesser than EECRP, and 29% lesser than EECA.

The energy conservation of the sensor nodes and the elongated lifetime of the network is the main objective of WSNs. As discussed above, existing algorithms have a lot of limitations. The PECR algorithm focuses on the crucial points and solves these problems competitively. It advances the state of the art by overpowering the limits of the existing work with the combination of an energy consumption model, suitable path selection, and CH and MCH functionalities. The designed scheme ultimately elongates the network's lifetime by achieving the goal in a significant way and gives an overall 44% improved performance over existing algorithms. Due to the improved performance, the PECR algorithm is optimum for real applications like the Internet of Things, smart applications, and numerous self-designed, independent applications.

Pros and Cons of the Proposed Study

The simulation results of the PECR algorithm are compared with standard and recent novel algorithms. The comparison shows its competitive behavior and significant improvement in terms of energy management. The proposed PECR algorithm has advantages over existing studies due to its energy consumption model and MCH. The blend of the PECR scheme reduces the traffic overhead at the BS, reserves energy, and ultimately extends the network lifetime. In cases of huge traffic, the PECR algorithm can create a burden on the MCH to some extent. It can be improved in the future by adding more than one MCH or a moveable BS.

5. Conclusions

In the WSN-assisted IoT, the productive utilization of restricted energy assets is an essential issue. In the field, low-power, static sensor nodes are deployed. After their arrangement, their substitution and recharge are troublesome and costly interactions. Therefore, the effective utilization of restricted energy resources is important to keeping a network practical for a more extended timeframe. To accomplish that goal, this study proposed the PECR algorithm, which is the blend of cluster arrangement, choosing CHs and an MCH, optimal path determination, the use of an energy utilization model during transmissions, MCH and CH alternation, and dead nodes' declaration assisted with lessening the traffic burden to avoid long distance transmission, improve energy effectiveness, and elongate the network life. The simulation results indicate that the PECR algorithm gives 44% improved performance over LEACH, EC, EECRP, and EECA. It is suitable where an extended lifetime

is needed. Because of the outcomes, it is a significant upgrade in WSN routing algorithms. In the future, additional work will be undertaken on the outcomes by fostering a more energy-proficient starting clustering strategy, the use of more than one MCH and BS, and portable circumstances to make it more realistic.

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